Chloaelis

A dual-channel filter, tube and delay instrument with a Lorenz attractor, a flock of boids and two field recordings built in.

Manual for version 1.1.0



About this manual

This manual covers Chloaelis 1.1.0, the plug-in (AU, VST3, LV2) and the desktop app. It follows the README, with the full key to the symbols, a reference for every parameter your DAW can automate, and answers to common problems.



Spatial audio version: Chloaeilius. Chloaelis as an Ambisonics encoder (1st to 7th order, 16 channels at 3rd order up to 64 at 7th, ACN/SN3D, for the IEM and SPARTA decoders): each side's dry sound is a point, its main echo rides the Lorenz attractor and its poly taps ride a 3-D flock, so the echoes fly round you, with distance and Doppler and a sphere you can turn to see where everything is. It lives in the



`Chloaeilius` folder of this repository.

What it is

Chloaelis is two complete signal paths, hard-panned: the **left input goes through the left side to the left output**, and the right through the right. Each side runs

input > multimode filter > tube > delay > output

(or delay > tube > filter, with the order switch). Shared controls in the middle tie the two sides together:

- **Filter**: continuous low pass > band pass > high pass > notch, with resonance.
- **Tube**: four types, each oversampled 4x: I 12AX7 (the default: warm, two triode stages), II EL84 (push-pull pentodes, crunchy), III 6L6 (with power-supply sag, it blooms) and IV starved (a low-voltage 12AX7, gated and sputtery).
- **Delay engines**: *tape* (an ageing tape loop: hiss, wow and flutter, oxide wear; it ages while it runs and *refresh* puts on new tape), *PT2399* (the delay chip, gritty at long times) and *BBD* (bucket brigade, dark and crunchy). Up to 1 s, regen that self-oscillates, a feedback high-pass, freeze, random freeze, and slew (pitch-bending) or fade time changes. The tape's hiss never cuts in or out: it fades up when you switch to tape and down when you leave it, so a recording has no step.
- **Karplus-Strong time** (ϙ): the delay becomes a tuned string. The time knob picks its note, snapped to one of 25 scales (the list from Cloudfin: La Monte Young's Well-Tuned Piano, ragas, maqamat, gamelan, Partch, Bohlen-Pierce, equal divisions), and regen is its sustain.
- **Poly**: fades in up to 8 delay taps per side at irregular offsets. All the way up it becomes a reverb-like cloud. **Flock** lets a boids simulation move those taps' times, so they drift, cluster and scatter.
- **Lorenz attractor**: its x and y move the left and right cutoffs.
- **LFOs**: triangle LFOs on filter mode, delay time and tube drive, per side.
- **Master feedback**, crossed over: the left output feeds the right input and the right output the left.
- **Two built-in recordings** to play through it: a long loop from *baserecording1* and a hydrophone recording. Fade them in and out; the left channel goes to the left side and the right to the right.
- Every switch (order, delay engine, time mode, tube type) **crossfades**: echoes ring out in the old setting while new sound goes through the new one, so there are no cuts.
- It **remembers** how you left it, and ↺ resets everything.
- **No words on its face**: every control is a letter, a mark or a picture (the **?** screen is the key). And two looks: the glass HUD, or **Brick** (▦), the Windows 2000 colour scheme in hard pixels.

Download and install

Ready-made builds for all three systems are on the **Releases page**:

System	File	Contains
macOS	`Chloaelis-macOS.zip`	the app, the Audio Unit and the VST3 (Intel + Apple Silicon)
Windows	`Chloaelis-Windows.zip`	the app (`Chloaelis.exe`) and the VST3
Linux	`Chloaelis-Linux.tar.gz`	the app, the VST3 and the LV2

`SHA256SUMS.txt` on the same page lists checksums if you want to verify a download.

> The builds aren't signed by Apple or Microsoft (that costs money), so the first launch needs one extra step on macOS and Windows. It's described below.

macOS (Homebrew)

If you use Homebrew:

```
brew install --cask femifleming/chloaelis/chloaelis
```

This installs the app into `/Applications` and the Audio Unit and VST3 into `~/Library/Audio/Plug-Ins`. To update later: `brew upgrade --cask chloaelis`. To remove: `brew uninstall --cask chloaelis`.

macOS (manual)

1. Download `Chloaelis-macOS.zip` from the Releases page and unzip it.
2. Drag **Chloaelis.app** into **Applications**.
3. For the plug-ins, open Finder, press ⇧⌘**G** and go to `~/Library/Audio/Plug-Ins`. Copy **Chloaelis.component** into the `Components` folder and **Chloaelis.vst3** into `VST3` (create either folder if it's missing).
4. First launch: because the build isn't notarised, macOS will refuse to open it the first time. Either right-click the app and choose **Open**, then **Open** again, or run this once in Terminal: ```sh xattr -dr com.apple.quarantine /Applications/Chloaelis.app ~/Library/Audio/Plug-Ins/Components/Chloaelis.component ~/Library/Audio/Plug-Ins/VST3/Chloaelis.vst3 ```
5. Restart your DAW (Logic, Ableton Live, Reaper and so on) so it rescans its plug-ins. Chloaelis appears under the maker **Chloaelis**.

Windows

1. Download `Chloaelis-Windows.zip` and unzip it.
2. **App**: put `Chloaelis.exe` anywhere and double-click it. If SmartScreen says "Windows protected your PC", click **More info** > **Run anyway** (it only asks once).
3. **Plug-in**: copy the `Chloaelis.vst3` folder into `C:\Program Files\Common Files\VST3`.
4. Restart your DAW so it rescans.

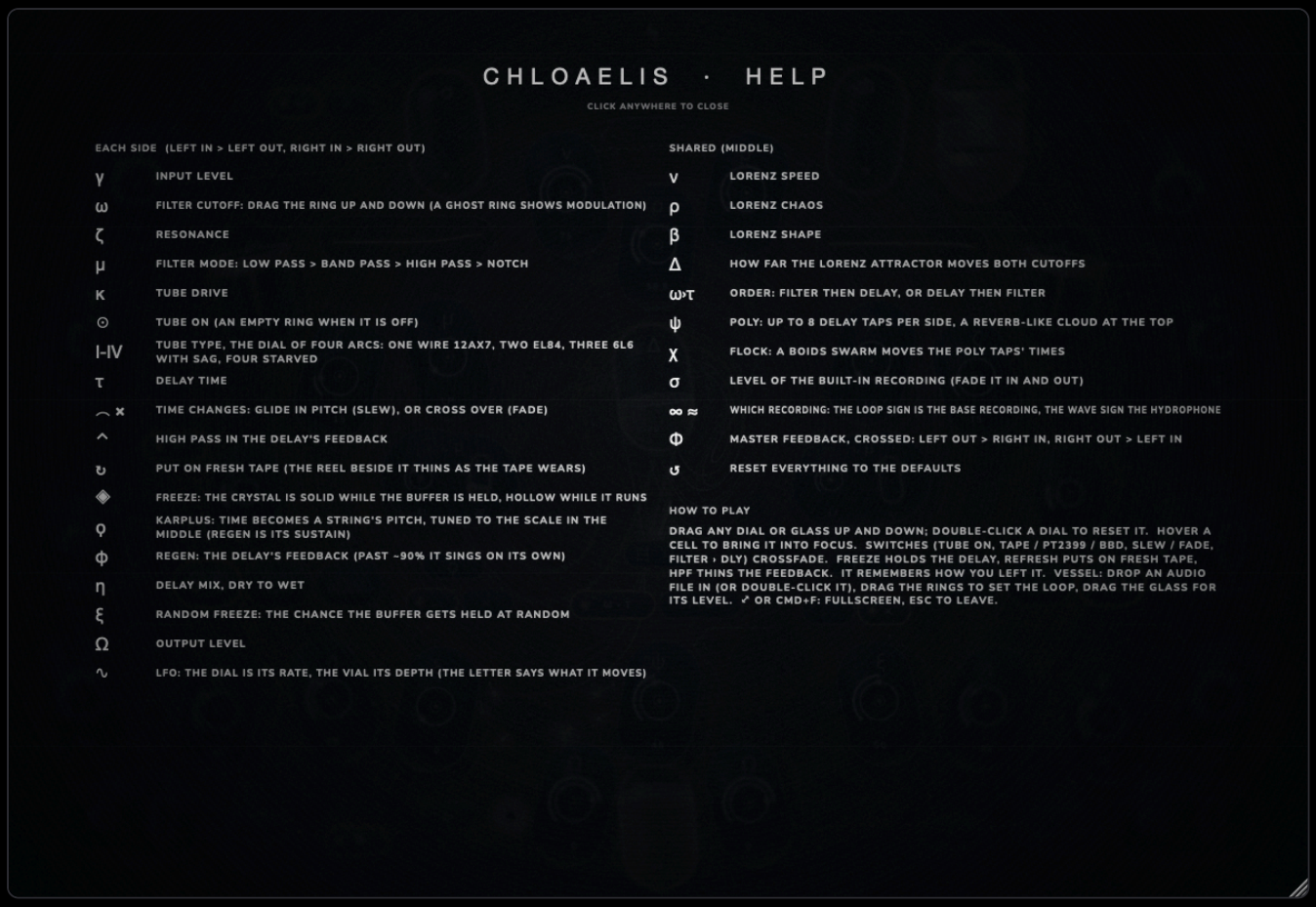
Linux

1. Download `Chloaelis-Linux.tar.gz` and unpack it: ```sh tar -xzf Chloaelis-Linux.tar.gz ```
2. **App**: run `./Chloaelis/Chloaelis` (you may want to copy it to `~/.local/bin`).
3. **Plug-ins**: ```sh mkdir -p ~/.vst3 ~/.lv2 cp -r Chloaelis/Chloaelis.vst3 ~/.vst3/ cp -r Chloaelis/Chloaelis.lv2 ~/.lv2/ ```

4. It needs ALSA or JACK, X11 and OpenGL, which a normal desktop already has.

How to use it

The layout



- The **left half** of the window is the left channel and the **right half** is the right channel. They mirror each other: input at the top, then filter, tube and LFOs, delay at the bottom, output at the bottom middle.
- The **middle** holds the shared controls: the Lorenz attractor (top), the poly and flock (bottom), the order switch, and the built-in recordings, master feedback and the vessel (top row).
- **Glowing dashed lines** show where signal and modulation flow. Pulses of light follow the signal.
- Cells **glow with level**: the side's colour when it's moving, warm white when it's hot and red when it's close to clipping.
- The **Lorenz attractor** floats in the background as a hologram, with dashed lines to the two cutoffs. The **boids** of the flock fly around the poly controls.

The symbols

There are no words on the face: every control is a Greek letter, a mark or a small picture. Press **?** (top right) for this list inside the app.

Each side

Symbol	Control	What it does
γ	Input	input level
ω	Cutoff	filter cutoff: drag the ring up and down. A faint ring shows where modulation has pushed it

ζ	Resonance	filter resonance
μ	Mode	filter mode: low pass > band pass > high pass > notch, continuously
κ	Drive	tube drive
the dial of four arcs	Tube type	the lit arc and the wires in its lens count the type: one 12AX7 (default), two EL84, three 6L6 with sag, four starved. Click for the next (alt- or right-click for the one before), or scroll
⊙ / ○	Tube on	switches the tube in and out: a lit heater, or an empty ring
τ	Time	delay time, 30 ms to 1 s
φ	Regen	the delay's feedback. Past about 90% it sings on its own
η	Mix	delay dry / wet
ξ	Random freeze	the chance that the delay buffer gets held for a while at random moments
reels / chip / buckets	Delay engine	three pictures: two reels (tape), a chip with bits going through it (PT2399), a row of buckets passing charge along (BBD). The one in use moves
⌒ / ×	Time changes	either glide (slew: pitch bends) or cross over (fade: no pitch change)
ϙ	Karplus-Strong time	the delay becomes a tuned string: see below
^	High pass	high-pass filter in the delay feedback
◈ / ◇	Freeze	holds the delay buffer. The crystal is solid while it is held (by hand or by the random freeze), hollow while it runs. On tape, a held loop slowly wears away
ʊ and the reel	Fresh tape	puts on fresh tape (tape engine only). The reel beside it is the tape's condition: its pack is fat when new and thins as the tape wears
Ω	Out	output level
∿	LFO	the dial is its rate, the small glass vial its depth. The letter next to ∿ says what it moves (μ mode, τ time, κ drive)

Shared (middle)

Symbol	Control	What it does
ν	Speed	how fast the Lorenz attractor moves
ρ	Chaos	the attractor's rho: low wanders gently, high is wild
β	Shape	the attractor's beta: reshapes the orbit
Δ	Depth	how far the attractor moves the two cutoffs (up to ±3.5 octaves)
ω › τ / τ › ω	Order	filter > tube > delay, or delay > tube > filter (the filter's and the delay's own letters, in order)
the ladder and a number	Scale	the scale the Karplus-Strong notes are tuned to, and how many notes it has. Hover to read its name, click for the list, or scroll through it
ψ	Poly	up to 8 delay taps per side. All the way up is a reverb-like cloud

χ	Flock	how much the boids move the poly taps' times (up to about ±0.9 octave)
∞ / ≈	Source	which built-in recording: ∞ the base loop, ≈ the hydrophone
σ	Source level	fades the built-in recording in and out
Φ	Master feedback	crossed: left out > right in, right out > left in
↺	Reset	every control back to its default (the tape's wear is kept)
▦	Brick	switches the look (top left): the glass HUD, or Brick
?	Help	the key to all of this, inside the app

Karplus-Strong time

ϙ (koppa, the old Greek K) on either side turns that side's delay into a Karplus-Strong string: a very short, exactly tuned delay that rings at a pitch.

- The **time knob (τ)** now picks a **note**. It still runs the same way round (up is longer, so lower): six octaves, from 27.5 Hz at the top of the knob to 1760 Hz at the bottom, and its readout shows the pitch in Hz. Notes snap to the chosen scale, whose 1/1 is A.
- The **scale** is the small cell in the middle (a ladder and the number of notes in the scale). Hover over it to read the name, click it for the list, or scroll. They are the scales of Cloudfin: The Well-Tuned Piano (La Monte Young), twelve ragas, 22 shruti, three maqamat, slendro and pelog, the harmonic series, Partch's 43 tones, Bohlen-Pierce, and 24-, 19-, 31- and 12-EDO.
- **Regen (φ)** is the string's sustain: most of its travel is long rings, and past 100% it sings on its own. **Freeze** holds the string.
- **Poly (ψ)** adds strings on other degrees of the scale, stacked up from the note you are on, so it plays a chord. **Flock (χ)** only detunes them a little.
- The **engine** pictures set the string's colour here: PT2399 bright, tape softer, BBD dark.
- **Slew (⌒)** glides between notes and **fade (×)** steps to them. The time LFO walks through the scale.

The two looks

▦ (top left) switches between the glass HUD and **Brick**: the Windows 2000 "Brick" colour scheme, with bevelled capsules, sunk white dials, progress-bar blocks in the vials and a maroon graph for the filter, all in hard pixels with no blur, glow or drift. The choice is kept with the state.

Playing it

- **Drag up and down** on any dial, glass vial or ring to change it. **Double-click** a dial to reset it to its default.
- **Hover** over a control to bring it into focus. The depth of field follows you, and the field tilts slightly toward the cursor.
- **No input?** Turn up **σ** to play one of the built-in recordings through it, and pick which with **∞ / ≈**.
- **Ideas to start with**
  - *Ambient wash*: σ up, ψ (poly) and χ (flock) up, φ (regen) around 60%, tape engine.
  - *Lo-fi echo*: PT2399 or BBD, τ around 400 ms, φ around 70%, κ (drive) up a little.
  - *Moving filter*: Δ (Lorenz depth) up, ρ (chaos) up, ζ (resonance) up.
  - *Self-playing*: Φ (master feedback) towards 90% with some regen: the two sides feed each other and keep going on their own.

    - *Tuned strings*: ϙ on, φ (regen) around 90%, ψ (poly) up for a chord, and move τ or its LFO to play the scale.

- Chloaelis **remembers its last state**: a new instance starts how the last one was closed. In a DAW, the project's own saved settings take priority.

The desktop app

The app (`Chloaelis.app`, `Chloaelis.exe` or `Chloaelis` on Linux) has a few extras:

- **Audio settings**: the **Options** button at the top left of the window chooses the audio device, the input and output channels and the buffer size.
- **Fullscreen**: the ⤢ button, or **⌘F** (Ctrl+F on Windows and Linux). It fills the whole screen with no menu bar or title bar, at any screen shape. **Esc** leaves it.
- **The vessel** (top right): a glass you can pour your own audio into.
    - **Drop an audio file** onto it (WAV, AIFF, FLAC, MP3, M4A, OGG or CAF, up to 10 minutes), or **double-click** it to choose one.
    - Its waveform runs up the inside, start at the bottom.
    - The **two rings** are the loop start and end: drag them.
    - **Drag the glass** itself (away from the rings) to set its level, shown by the liquid.
    - It plays into both sides like the built-in recordings, and is remembered with the rest of the state.

Help inside the app

Press **?** (top right) for a full key to the symbols and a short how-to. Click anywhere, or press Esc in the app, to close it.

Build from source

You need CMake 3.22 or newer and a C++20 compiler (Xcode on macOS, Visual Studio 2022 on Windows, GCC or Clang on Linux).

```
git clone --recursive https://github.com/femifleming/Chloaelis.git
cd Chloaelis
cmake -B build -DCMAKE_BUILD_TYPE=Release
cmake --build build --config Release --target Chloaelis_All
```

The results are in `build/Chloaelis_artefacts/Release/` (`Standalone`, `VST3`, plus `AU` on macOS and `LV2` on Linux). Add `-DCHLOAELIS_COPY_AFTER_BUILD=ON` to have the plug-ins copied into your user plug-in folders after each build.

On Linux, install the development packages first (Debian / Ubuntu names):

```
sudo apt-get install libasound2-dev libjack-jackd2-dev libfreetype-dev libfontconfig1-dev \
  libx11-dev libxcomposite-dev libxcursor-dev libxext-dev libxinerama-dev libxrandr-dev \
  libxrender-dev libxi-dev libegl-dev libgl1-mesa-dev libglu1-mesa-dev mesa-common-dev ladspa-
sdk
```

If you cloned without `--recursive`, run `git submodule update --init` to fetch JUCE.

Tests (macOS): `cmake -B build -DCHLOAELIS_BUILD_TESTS=ON` builds `ChloaelisTest` (offline DSP checks) and `ChloaelisHostTest` (the full plug-in, headless; `CHL_AUDIOTEST=1` runs the audio checks).

Human contributions

Chloaelis's code was written with Claude Code (Anthropic), working under the direction of its author, femifleming. This is what the author brought to it:

- **The concept and the signal path.** Two of Bizarre Jezabel's *Chloe* Eurorack modules, hard-panned left and right, each running filter > tube distortion > PT2399 delay. Includes a buffer freeze, with the PT2399's artefacts kept in.
- **The name.** Chloaelis was its own plug-in from the start, separate from the author's other plug-ins.
- **The three delay engines and their sound references.** Tape, PT2399 and BBD. The tape has hiss, about 2 % warble, wear that builds as it runs and a refresh button, after clippingrainbirds.world, William Basinski, Ian William Craig and Amulets. The BBD is crunchy, like the Serge delay.
- **The modulation ideas.**
  - A Lorenz attractor with one output to each cutoff.
  - Random freeze.
  - Slew vs fade time changes.
  - A delay/filter order switch.
  - Poly taps: offset copies, up to 8 per side.
  - A boids "flock" knob for tap chaos.
  - Crossed master feedback.
- **What was cut.** The vactrol/LPG/VCA section, the per-parameter LFOs, the VU meters (colour shows level instead), the black pop-ups and the background text. Tube types came back with the original as the default, and every switch crossfades instead of cutting.
- **The visual direction.** It comes from the 2012 game *Splice*: a network of cells, dark glow, grain, bokeh and a tilt-shift focus that follows the cursor. The author toned the perspective down several times and turned down other looks along the way.
- **Interface details.**
  - HUD dials with one thin continuous ring.
  - Glass capsule cells.
  - Greek-letter labels instead of text.
  - Glass vials only for depth, and sliding collars for the cutoffs.
  - Colour that opens cool and grey and warms with the input.
  - A plain gradient icon.
- **The built-in recordings.** Two of the author's own field recordings ship in the plug-in: a base recording and a hydrophone recording.
- **The desktop app.** The drag-in sample vessel, true fullscreen on Cmd+F, remembering the last state, and a reset button.
- **Testing.** The author reported lag (and set a 30 fps minimum), pixelation, aliasing in the blur, stepped focus and hover jitter, and had them fixed.
- **Version 1.1.** The author asked for the tape hiss to fade rather than cut, for Karplus-Strong time tuned to Cloudfin's scales, for a Windows 2000 look (chosen as *Brick* from a set of mock-ups), for symbols in place of every word, and for the cells to sit closer together; and picked the tube dial, the engine pictures, the crystal for freeze and the reel for the tape from options drawn for them.
- **The release.** Builds for macOS, Windows and Linux, a README, in-app help and a Homebrew install, with credit to *Splice* and *Chloe*.

Credits and licence

- Design and recordings: **Femi Fleming**. The two built-in recordings are Femi Fleming's field recordings (*baserecording1* and *hydrophone, June 2026, 3*).
- Inspired by the 2012 game *Splice* and the *Chloe* Eurorack module by Bizarre Jezabel.
- Built with JUCE 9.0.5, used under the GNU AGPLv3.
- Interface font: Nunito (a bold cut), SIL Open Font License 1.1 (see `Resources/fonts/OFL-Nunito.txt`).

Chloaelis is free software: you can redistribute it and/or modify it under the terms of the **GNU Affero General Public License v3** as published by the Free Software Foundation. See LICENSE.

Parameter reference

Every control your DAW can automate, by the name it shows there. The two sides have the same controls, so the right side's table follows the left's. Defaults are what a fresh instance starts with (reset brings them back).

Shared (the middle)

Name in your DAW	ID	Range	Default
Lorenz Speed	`lz_speed`	0% to 100%	35%
Lorenz Chaos	`lz_chaos`	ρ 24.5 to ρ 180.0	ρ 38.5
Lorenz Shape	`lz_shape`	β 0.50 to β 4.00	β 2.67
Lorenz > Cutoff	`lz_depth`	0% to 100%	0%
Poly Width	`width`	OFF to 100%	OFF
Poly Flock	`poly_flock`	0% to 100%	0%
Order	`order`	Filter > Delay, Delay > Filter	Filter > Delay
Source	`src_sel`	Base Loop, Hydrophone	Base Loop
Source Level	`src_level`	0% to 100%	0%
Master Feedback	`master_fb`	0% to 100%	0%
Karplus-Strong Scale	`ks_scale`	The Well-Tuned Piano (La Monte Young), Raga Bhairav, Raga Yaman, Raga Bhairavi, Raga Todi, Raga Marwa, Raga Purvi, Raga Kafi, Raga Malkauns, Raga Darbari Kanada, Raga Bhupali, Raga Desh, Raga Puriya Dhanashri, 22 Shruti, Maqam Rast, Maqam Bayati, Maqam Saba, Slendro, Pelog, Harmonic Series 8-16, Partch 43-Tone, Bohlen-Pierce (tritave), Quarter-tone (24-EDO), 19-EDO, 31-EDO, Chromatic (12-EDO)	The Well-Tuned Piano (La Monte Young)

Left side

Name in your DAW	ID	Range	Default
L In Att	`l_inatt`	silent to 12.0 dB	0.0 dB
L Cutoff	`l_cutoff`	20 Hz to 20.0 kHz	20.0 kHz
L Resonance	`l_res`	0% to 100%	15%
L Mode	`l_morph`	LP to Notch	LP
L Drive	`l_drive`	0% to 100%	30%
L Time	`l_time`	30 ms to 1000 ms	330 ms
L Regen	`l_regen`	0% to 115%	40%
L Wet	`l_wet`	0% to 100%	35%
L Out	`l_out`	silent to 12.0 dB	0.0 dB

L Random Freeze	l_randfrz	0% to 100%	0%
L LFO>Mode Rate	l_lfomode_rate	0.02 Hz to 20.0 Hz	0.21 Hz
L LFO>Mode Depth	l_lfomode_depth	0% to 100%	0%
L LFO>Time Rate	l_lfotime_rate	0.02 Hz to 20.0 Hz	0.11 Hz
L LFO>Time Depth	l_lfotime_depth	0% to 100%	0%
L LFO>Drive Rate	l_lfodrive_rate	0.02 Hz to 20.0 Hz	0.37 Hz
L LFO>Drive Depth	l_lfodrive_depth	0% to 100%	0%
L Delay HPF	l_hpf	off / on	Off
L Freeze	l_freeze	off / on	Off
L Tube On	l_tubeon	off / on	On
L Delay Type	l_dtype	Tape, PT2399, BBD	PT2399
L Time Change	l_tmode	Slew, Fade	Slew
L Tube Type	l_tubetype	12AX7, EL84, 6L6, Starved	12AX7
L Karplus-Strong Time	l_ks	off / on	Off

Right side

Name in your DAW	ID	Range	Default
R In Att	r_inatt	silent to 12.0 dB	0.0 dB
R Cutoff	r_cutoff	20 Hz to 20.0 kHz	20.0 kHz
R Resonance	r_res	0% to 100%	15%
R Mode	r_morph	LP to Notch	LP
R Drive	r_drive	0% to 100%	30%
R Time	r_time	30 ms to 1000 ms	410 ms
R Regen	r_regen	0% to 115%	40%
R Wet	r_wet	0% to 100%	35%
R Out	r_out	silent to 12.0 dB	0.0 dB
R Random Freeze	r_randfrz	0% to 100%	0%
R LFO>Mode Rate	r_lfomode_rate	0.02 Hz to 20.0 Hz	0.17 Hz
R LFO>Mode Depth	r_lfomode_depth	0% to 100%	0%
R LFO>Time Rate	r_lfotime_rate	0.02 Hz to 20.0 Hz	0.13 Hz
R LFO>Time Depth	r_lfotime_depth	0% to 100%	0%
R LFO>Drive Rate	r_lfodrive_rate	0.02 Hz to 20.0 Hz	0.29 Hz
R LFO>Drive Depth	r_lfodrive_depth	0% to 100%	0%
R Delay HPF	r_hpf	off / on	Off
R Freeze	r_freeze	off / on	Off
R Tube On	r_tubeon	off / on	On
R Delay Type	r_dtype	Tape, PT2399, BBD	PT2399
R Time Change	r_tmode	Slew, Fade	Slew
R Tube Type	r_tubetype	12AX7, EL84, 6L6, Starved	12AX7

R Karplus-Strong Time	r_ks	off / on	Off

Problems and answers

macOS says it can't be opened, or the DAW doesn't list it

The builds aren't notarised. Run `xattr -dr com.apple.quarantine` on the app and the plug-ins (see Installing), then rescan in your DAW.

There's no sound and nothing on the input

Turn up σ (source level) to play a built-in recording through it, and pick which with ∞ / ≈. In the desktop app, check Options for the audio device.

It keeps sounding after I stop

Regen past about 90 % sings on its own, and master feedback (Φ) keeps the two sides feeding each other. Turn them down, or reset with ↺.

A new instance doesn't start from the defaults

It remembers its last state, so a new instance starts how the last one was closed. A saved project still opens as you saved it. ↺ resets everything (the tape's wear is kept).